KENYA MEDICAL TRAINING COLLEGE
FACULTY OF CLINICAL SCIENCES
DEPARTMENT OF ORTHOPAEDIC AND TRAUMA MEDICINE

FINAL QUALIFYING EXAMINATION
FOR
DIPLOMA IN ORTHOPAEDIC AND TRAUMA MEDICINE

PAPER: ORTHOPAEDICS

DATE: 4TH FEBRUARY, 2025 **TIME: 3 HOURS (9:00 – 12:00 NOON)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answers Question)
2. Attempt **ALL** Questions
3. **ALL** Multiple choice questions **MUST** be answered in **CAPITAL LETTERS**
4. Write your **EXAMINATION NUMBER** on all the answer sheets provided and on the question paper.
5. No other writing should be done on the question paper except your Examination Number.
6. Ensure that all examination answer scripts are handed in at the end of the examination
7. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

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SECTION 1: MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. A 10-year-old boy with history of mild trauma in the last three days, presents with severe pain and swelling in his leg just below the knee. On examination he is febrile and you elicit marked tenderness on palpation. Which of the following is the best diagnostic test for this case?
   A. CT scan
   B. Blood culture
   C. Full haemogram
   D. Plain x-ray
2. you are viewing an x-ray film of a patient who has chronic osteomyelitis and you identify a sequestrum. Your conclusion is based on which of the following radiographic appearance?
   A. Radio-opaque
   B. Highly radio opaque
   C. Hypo dense
   D. Radiolucent
3. An orthopaedic condition that presents with brittle bones and at times bluish sclera is: -
   A. Osteoporosis
   B. Osteogenesis imperfecta
   C. Osteomalacia
   D. Osteopetrosis
4. Multiple bone fractures in newborns are seen in:
   A. Syphilis
   B. Osteogenesis imperfecta
   C. Scurvy
   D. Osteomalacia
5. Which one of the following is a radiological feature in chronic osteomyelitis?
   A. Onion peel appearance
   B. Lamellar bone
   C. Involucrum
   D. Osteophytes
6. The most common source of bone and joint infection is:
   A. Direct spread
   B. Percutaneous
   C. Lymphatic
   D. Haemotogenous
7. A long standing localized pyogenic abscess in the bone is known as?
   A. Brodie's abscess
   B. Cold abscess
   C. Sub periosteal abscess
   D. Soft tissue abscess

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8. Which of the following is a malignant tumor?
   A. Chondrosarcoma
   B. Chondroma
   C. Oesteochondroma
   D. Enchondroma
9. The most common site of eosinophilic granuloma is?
   A. Femur
   B. Lumbar vertebra
   C. Radius
   D. Skull
10. A cause of painful hip in an infant is:
   A. Infantile coxa vera
   B. Poliomyelitis
   C. Congenital dislocation of the lip
   D. Transient synovitis
11. Bony ankylosis results from?
   A. Tuberculosis arthritis
   B. Pyogenic arthritis
   C. Osteoarthritis
   D. Rheumatoid arthritis
12. A patient with tertiary syphilis most frequently presents with arthritis of?
   A. Shoulder
   B. Elbow joint
   C. Knee joint
   D. Wrist joint
13. Tuberculosis of the spine starts in
   A. Vertebral body
   B. Nucleous pulposus
   C. Paravertebral fascia
   D. Annulus
14. The most common cause of genu valgum in children is?
   A. Rickets
   B. Rheumatoid arthritis
   C. Osteoarthritis
   D. Paget's diseases
15. Cold abscess is commonly due to the infection of
   A. Ribs
   B. Spine
   C. Skull
   D. Sternum

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16. Diffuse joint swelling can be caused by
    A. Fluid within the joint
    B. Joint dislocation
    C. Excessing joint moment
    D. baker's cyst
17. Arthrodesis is indicated in
    A. Advanced osteoarthritis
    B. Osteomyelitis
    C. Joint dislocation
    D. Osteomalacia
18. Drug of choice in senile osteoporosis is
    A. Androgen
    B. Estrogen
    C. Calcitonin
    D. Ethidronate
19. Which of the following is the action of vitamin D?
    A. Stimulate bone marrow
    B. Increase calcium loss
    C. Stimulates absorption of calcium
    D. Stimulates osteoclast activity
20. Calcitonin is used in the treatment of?
    A. post-menopausal osteoporosis
    B. malignant hypercalcemia
    C. osteomalacia
    D. Paget's disease
21. Sprengel's shoulder is due to deformity of
    A. Scapula
    B. Humerus
    C. Clavicle
    D. Vertebra
22. The maneuver that test congenital hip dysplasia by reducing a dislocated hip is referred as:
    A. Barlow's test
    B. Telescoping
    C. Trendelenburg test
    D. Ortolani test
23. Rocker bottom foot is due to
    A. Over corrected club foot
    B. Poliomyelitis
    C. Vertical talus
    D. Deformities of spine

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24. Which muscle is responsible for initiation of abduction at the shoulder joint?
    A. Teres major
    B. Supraspinatus
    C. Pectoralis major
    D. Deltoid muscle
25. Osteogenesis imperfecta is:
    A. Autosomal dominant (AD)
    B. Autosomal recessive (AR)
    C. Both AD & AR
    D. Sex linked dominant
26. Club foot is common among?
    A. Females
    B. Males
    C. Uni-ovular twins
    D. Bi-novular twins
27. Metaphyseal lesion in children includes:
    A. Osteoporosis
    B. Osteomyelitis
    C. Osteomalacia
    D. Osteoid osteoma
28. A 56-year-old female presents with headache, chronic back pain and generalized body malaise. Radiograph shows multiple punched out osteolytic lesions on the skull vault. These findings are seen in:
    A. Hyperparathyroidism
    B. Multiple myeloma
    C. Renal osteodystrophy
    D. Hypoparathyroidism
29. Which of the following is a cause of osteitis fibrosa cystica?
    A. Paget's disease
    B. Polyostotic fibrous dysplasia
    C. Parathyroid adenoma
    D. fibrochromatosis
30. Which one of the following is a laboratory finding in osteomalacia?
    A. Low alkaline phosphatase
    B. Raised Serum phosphate
    C. Raised serum calcium
    D. Raised alkaline phosphatase
31. Which one of the following bones is spared in bone resorption associated with hyperparathyroidism?
    A. Jaws
    B. Metacarpals
    C. Ribs
    D. End of long bones

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32. The most common cause of bilateral genu valgus is
   A. Tuberculosis
   B. Multiple myeloma
   C. Osteoporosis
   D. Rickets
33. A patient presents with multiple joint pains for one year. On examination you find swan neck deformity of the fingers. This is a feature of?
   A. Syphilitic arthritis
   B. Gouty arthritis
   C. Osteoarthritis
   D. Rheumatoid arthritis
34. Charcot's joints are associated with?
   A. Gouty arthritis
   B. Syphilis
   C. Tuberculosis
   D. Rheumatoid arthritis
35. Acute osteomyelites of the long bones commonly affects
   A. Metaphysis
   B. Diaphysis
   C. Epiphysis
   D. Articular surface
36. Which of the following is affected first in rheumatoid arthritis?
   A. Synovial fluid
   B. Sub chondral bone
   C. Synovial membrane
   D. Cartilage
37. Which of the following joints is commonly affected by ankylosing spondylitis?
   A. Ankle
   B. Knee
   C. Shoulder
   D. Wrist
38. Which one of the following drugs is used in an acute attack of gouty arthritis?
   A. Colchicine
   B. Allopurinol
   C. Tramadol
   D. Probenecid
39. Avascular necrosis of the head by the femur can occur in
   A. Sickle cell anaemia
   B. Caisson's disease
   C. Extracapsular fracture neck of femur
   D. Trochanteric fracture femur

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40. Enchondroma should be treated by
    A. Irradiation and chemotherapy
    B. Amputation and radiotherapy
    C. Local excision and chemotherapy
    D. Curettage and bone chips grafting

SECTION 2: SHORT ANSWER QUESTIONS (40 MARKS)

41. A patient has been diagnosed with Paget's disease. Elaborate five clinical features of this condition **(5marks)**
42. State five hormones involved in bone homeostasis **(5marks)**
43. A 60-year-old female presents with fragility fracture of the proximal humerus. Outline five predisposing factors for the above condition **(5marks)**
44. Distinguish between osteoarthritis and rheumatoid arthritis. **(5marks)**
45. A patient with chronic hip pain is diagnosed with avascular necrosis of the head of femur. Elaborate five (5) predisposing factors to this condition. **(5marks)**
46. A. newborn is brought to orthopaedic clinic with bilateral foot deformity which is identified as CTEV. Describe the conservative treatment for this child **(5marks)**
47. A 50-year-old male who has been taking alcohol and roasted beef, presents with severe pain and swelling in the big toe. Lab investigations shows elevated serum uric acid levels State five complications of this condition **(5marks)**
48. 5-year child presents with history of knee joint pain, swelling and fever. On examination the joint is warm, tender and fluctuant. Diagnose and treat this patient **(5marks)**

SECTION 3: LONG ANSWER QUESTION (20 MARKS)

49. 15-year-old boy presents pain and a mass around the knee joint. You make a diagnosis of osteogenic sarcoma. discuss the radiological finding, staging and treatment of this patient **(20 marks)**

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